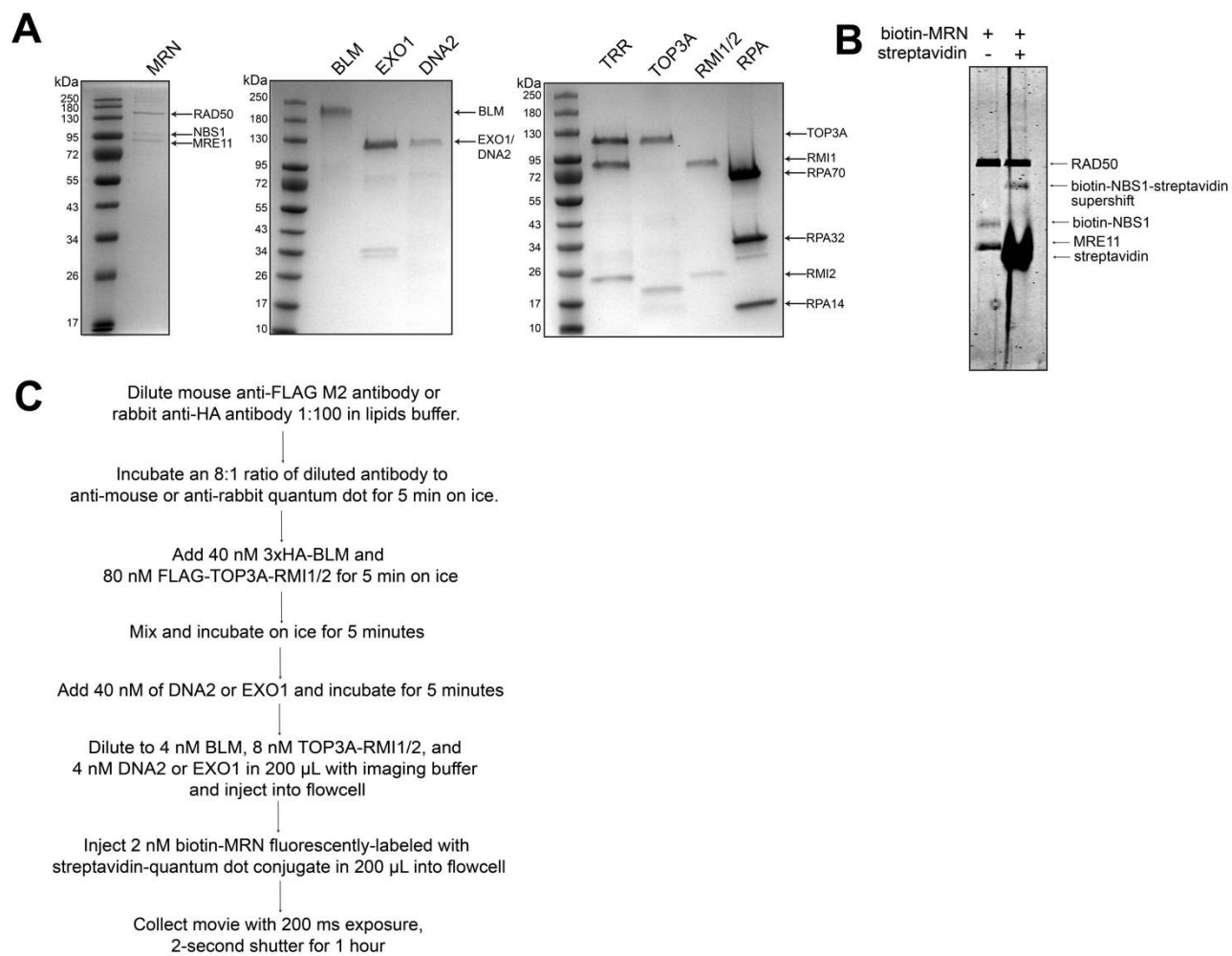
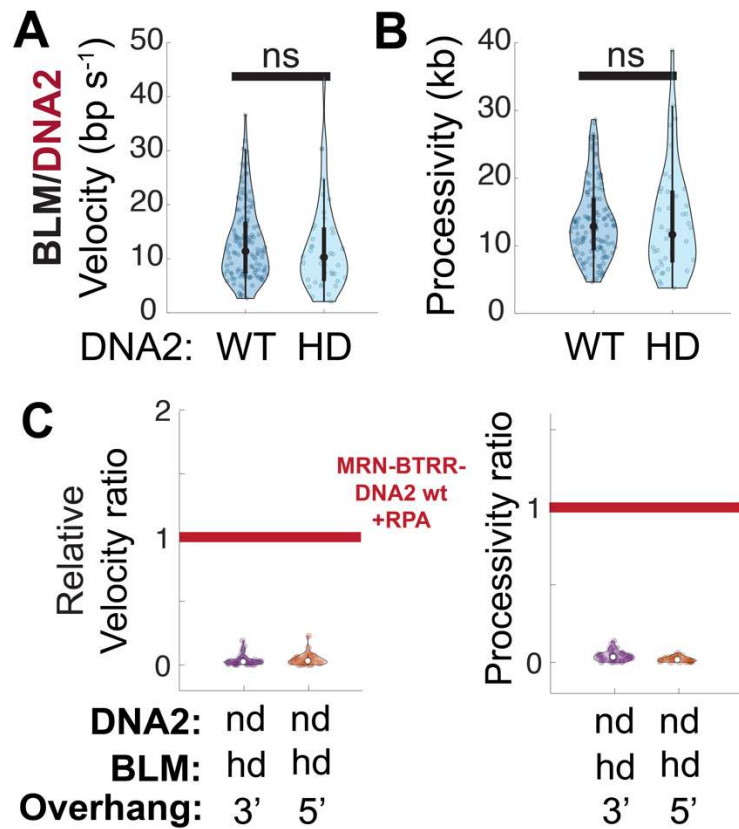


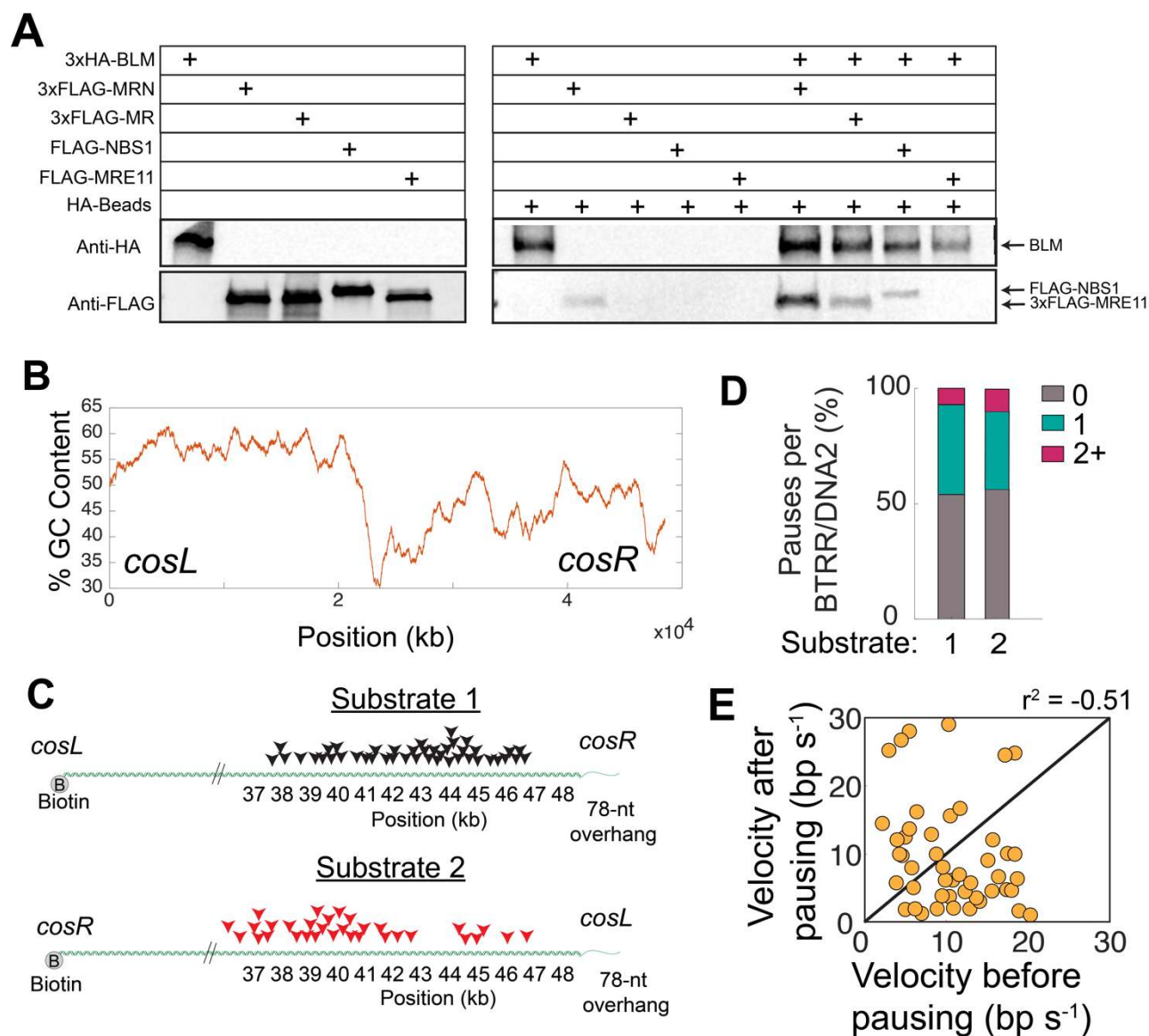
SUPPLEMENTAL MATERIAL



Supplemental Figure 1. (A) SDS-PAGE gels of the recombinant proteins used in this study. **(B)** SDS-PAGE gel showing biotin-MRN and biotin-MRN + streptavidin. The samples were not boiled to preserve the biotin-streptavidin interactions. The upshifted biotin-NBS1-streptavidin conjugates are indicated. The complete disappearance of the biotin-NBS1 band indicates that nearly 100% of the purified MRN is biotinylated. **(C)** Flowchart describing the single-molecule resection experiments.



Supplemental Figure 2. (A) Velocities and (B) processivities of the wild type (WT) BLM/DNA2 (n=126), and the helicase-deficient (HD) DNA2(K654R) mutant (n=42). Black bars show the interquartile range (thick bars) and 1.5x interquartile range (thin bars). The black dot in the middle is the median (C) Ratio of MRN/BTRR/DNA2 velocities (left) and processivities (right) with nuclease-deficient (nd) DNA2 and helicase-deficient (hd) BLM mutants (n>30 for both conditions from two flowcells). Both velocity and processivity are normalized to the corresponding values for the WT MRN/BTRR/DNA2 complex (red line). (not significant; ns, p>0.05).



Supplemental Figure 3. (A) *In vitro* pulldown assays show that BLM interacts with NBS1 and RAD50. **(B)** The DNA substrate GC content is higher near *cosL* relative to *cosR*. The substrate is derived from λ -phage DNA. The **(C)** Pausing positions for both DNA substrates (n=42 and 35 pauses for substrates 1 and 2, respectively). **(D)** Pausing frequency per resectosome for the indicated substrates. **(E)** Velocities of individual BTRR/DNA2 complexes before and after pausing. Dashed line is shown as a reference with a slope of $m = 1$ (n=42 resectosomes).

Table S1: Velocity and processivity for DNA2 and EXO1-mediated DNA resection

<u>Sample</u>	<u>Processivity (kb)</u> mean $\pm$ st. dev.	<u>Velocity (bp s⁻¹)</u> mean $\pm$ st. dev.	<u>Number of molecules (n)</u>
BLM/DNA2	13 $\pm$ 6	13 $\pm$ 7	126
MRN/BLM/DNA2	12 $\pm$ 5	9 $\pm$ 6	30
BTRR/DNA2	13 $\pm$ 5	9 $\pm$ 6	94
MRN/BTRR/DNA2	18 $\pm$ 6	18 $\pm$ 11	82
BLM/EXO1	15 $\pm$ 7	13 $\pm$ 9	124
MRN/BLM/EXO1	14 $\pm$ 6	12 $\pm$ 7	82
BTRR/EXO1	12 $\pm$ 8	14 $\pm$ 11	79
MRN/BTRR/EXO1	12 $\pm$ 7	13 $\pm$ 10	57
MRN/BTRR/DNA2 (D277A)	2 $\pm$ 2	2 $\pm$ 2	89
MRN/BTRR/DNA2 (K654R)	13 $\pm$ 6	11 $\pm$ 7	76
MRN/BLM (K695A)/TRR/DNA2 (D277A) + 3'-overhang	0.7 $\pm$ 0.6	0.7 $\pm$ 0.7	44
MRN/BLM (K695A)/TRR/DNA2 (D277A) + 5'-overhang	0.4 $\pm$ 0.3	0.8 $\pm$ 0.8	32
MR/BTRR/DNA2	6 $\pm$ 3	15 $\pm$ 12	42
Mre11/BTRR/ DNA2	10 $\pm$ 5	14 $\pm$ 10	63
MRN(S1202R)/BTRR/ DNA2	8 $\pm$ 4	17 $\pm$ 13	87
BLM/DNA2 HD	14 $\pm$ 8	12 $\pm$ 8	42

Table S2: Velocity and processivity for DNA2 helicase activity

<u>Sample</u>	<u>Processivity (kb)</u> mean $\pm$ st. dev[	<u>Velocity (bp s⁻¹)</u> mean $\pm$ st. dev[	<u>Number of molecules (n)</u>
DNA2 (D277A)	4 $\pm$ 2	5 $\pm$ 3	23
MRN/DNA2 (D277A)	3 $\pm$ 1	3 $\pm$ 1	19
TRR/DNA2 (D277A)	4 $\pm$ 2	3 $\pm$ 2	27
MRN/TRR/ DNA2 (D277A)	4 $\pm$ 1	4 $\pm$ 2	15

Table S3: Velocity and processivity for BLM helicase activity

<u>Sample</u>	<u>Processivity (kb)</u> mean $\pm$ st. dev.	<u>Velocity (bp s⁻¹)</u> mean $\pm$ st. dev.	<u>Number of molecules (n)</u>
BLM	17 $\pm$ 7	25 $\pm$ 18	90
BTRR	14 $\pm$ 8	14 $\pm$ 11	86
BLM/TOP3A	13 $\pm$ 8	13 $\pm$ 9	83
BLM/RMI1/2	8 $\pm$ 3	10 $\pm$ 6	34
BTRR/RPA	11 $\pm$ 6	17 $\pm$ 11	67
MRN/BTRR/RPA	14 $\pm$ 7	12 $\pm$ 8	131
MR/BTRR/RPA	8 $\pm$ 3	13 $\pm$ 7	68
Mre11/BTRR/RPA	6 $\pm$ 3	11 $\pm$ 7	66
MRN(S1202R)/BTRR/RPA	8 $\pm$ 5	17 $\pm$ 13	30

Table S4: Oligonucleotides used in this study.

Name	Sequence
IF006	[p]GGGCGGCGACCT[Bio]
IF007	[p]AGGTCGCCGCCC[Bio]
LM003	[p]GGGCGGCGACCT TT TT
LM024	[p]AGGTCGCCGCCC TT TT
MS0015	TATTTTGCCTTTCAATCCAAACCTAGG
MS0016	GCCGTTACAGTTGGTGTGAAAATACATCGAG
MS0017	TCCTGTCCCAGGCATACCC
MS0018	CGTACAACCTACGATATGTACTCTCGTAAGAATTC